°1.1

Найти: $\frac{F_э}{F_г}$

Решение:

$F_э = k\frac{q_1 q_2}{r^2}$ $F_г = G\frac{m_1 m_2}{r^2}$

$\frac{F_э}{F_г} = \frac{k}{G}\frac{q_p^2}{m_p^2} = \frac{9\cdot 10^9}{6{,}67\cdot 10^{-11}}\cdot(9{,}58\cdot 10^7)^2 = 1{,}23\cdot 10^{36}$

Ответ: $1{,}2\cdot 10^{36}$

°1.2

$E = 100$ В/м
$h = 1{,}5$ км
$E' = 25$ В/м

Найти: $n_э$

Решение:

$\frac{\partial E}{\partial x} = 4\pi n_э \cdot q \Rightarrow E - E' = 4\pi n_э \cdot q \cdot h$ $n_э = \frac{E-E'}{4\pi q \cdot h} = \frac{75\ \text{В/м}}{4\cdot\pi\cdot 1{,}6\cdot 10^{-19}\cdot 1{,}5\cdot 10^3} =$

$=$

1.7

$q; d; P$

Найти: F

Решение:

$\vec{E} = \frac{3(\vec{P}\cdot\vec{r})}{r^5}\vec{r} - \frac{\vec{P}}{r^3}$

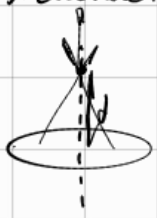
$\vec{F} = q\vec{E} = \frac{2\vec{P}}{d^3}q$ Ответ: $\frac{2\vec{P}}{d^3}q$

1.10

$R; \sigma; d$

Найти: E

Решение:



$\sigma = \frac{Q}{\pi R^2}$

$dE = \frac{dq}{d^2+r^2} = \frac{\sigma 2\pi r dr}{d^2+r^2}$

$E(d) = E_{\parallel d} = \int_0^R \cos\alpha \frac{2\pi r\sigma dr}{d^2+r^2} =$

$= \int_0^R \frac{d}{\sqrt{d^2+r^2}}\frac{2\pi r\sigma dr}{d^2+r^2} = 2\pi\sigma d\int_0^R \frac{r\,dr}{(d^2+r^2)^{3/2}} =$

$= 2\pi\sigma\left(1 - \frac{d}{\sqrt{r^2+d^2}}\right)$

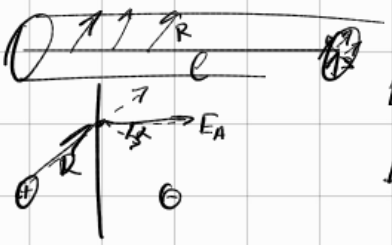
Ответ: $2\pi\sigma\left(1-\frac{d}{\sqrt{r^2+d^2}}\right)$

1.14

$d; +\varkappa; -\varkappa$

Найти: $E(h)$

Решение:



$\Phi = E\cdot S = E\cdot 2\pi r R = \frac{\varkappa\cdot l}{\varepsilon_0}$

$E\cdot 2\pi l = \frac{\varkappa\cdot l}{\varepsilon_0}$ $E = \frac{\varkappa}{2\pi\varepsilon_0 R} = \frac{2\varkappa}{R}$

$E_A = 2E\cdot\cos\alpha = \frac{4\varkappa}{R}\frac{d}{2R} = \frac{2\varkappa d}{(\frac{d}{2})^2+h^2} = \frac{2\varkappa d}{d^2+4h^2}$

Ответ: $\frac{8\varkappa d}{d^2+4h^2}$

Т1.1

Дано:

$P = 1{,}84\ D$

$r = 35\ Å$

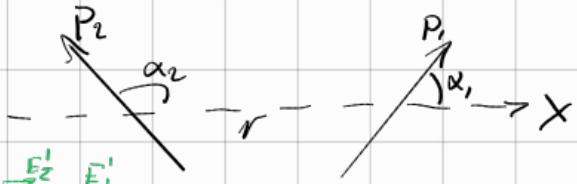
а) $\alpha_1 = \alpha_2 = 0$

б) $\alpha_1 = \frac{\pi}{2}\quad \alpha_2 = \pm\frac{\pi}{2}$

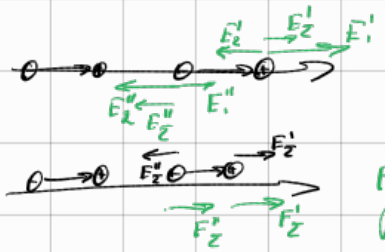
в) $\alpha_1 = 0\quad \alpha_2 = \pm\frac{\pi}{2}$

Найти: F; a_{max}

Решение



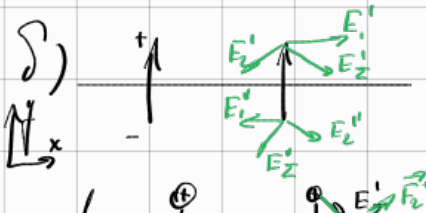
а)



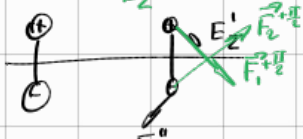
$F_y = 0$

$$F_x = P\frac{\partial E}{\partial x} = P\frac{\partial}{\partial x}\left(\frac{3P}{x^3} - \frac{P}{x^3}\right) = -\frac{6P^2}{x^4} = -14\cdot10^{-10}\ дин = -14\cdot10^{-15}\ Н$$

б)



1.

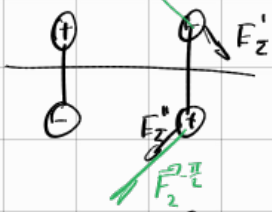


$F_y^{\pm\frac{\pi}{2}} = 0$

$F_x^{\pm\frac{\pi}{2}} \neq 0$

$P = ql$

2.



$F_y^{\pm\frac{\pi}{2}} = 0$

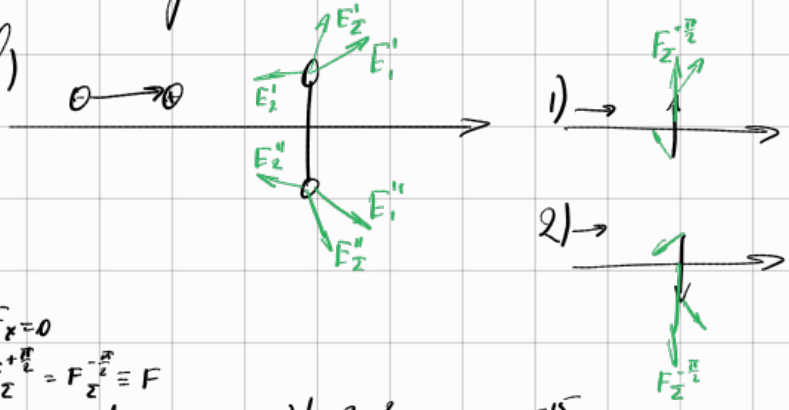
$F_x^{\pm\frac{\pi}{2}} \neq 0$

$\left|F_x^{-\frac{\pi}{2}}\right| = \left|F_x^{+\frac{\pi}{2}}\right| \equiv F_x$

$$|F_x| = \left|P\cdot\frac{\partial E}{\partial x}\right| = \left|P\frac{\partial}{\partial x}\left(-\frac{P}{x^3}\right)\right| = \frac{3P^2}{x^4} = \frac{3P^2}{r^4} = \frac{3\cdot(1{,}84\cdot10^{-18})^2}{(35\cdot10^{-2})^4} =$$

$$= 7\cdot10^{-10}\ дин = 7\cdot10^{-15}\ Н$$

в)



$F_x = 0$

$F_z^{+\frac{\pi}{2}} = F_z^{-\frac{\pi}{2}} \equiv F$

$$|F_x| = \left|P\frac{\partial E}{\partial x}\right| = \left|P\frac{\partial}{\partial x}\left(-\frac{P}{x^3}\right)\right| = \frac{3P^2}{x^4} = 7\cdot10^{-15}\ Н$$

$$a_{max} = \frac{F}{m} = \frac{7\cdot10^{-15}\cdot6\cdot10^{23}}{18\cdot10^{-3}} \simeq 2{,}3\cdot10^{11}\frac{1}{c^2} = 2{,}3\cdot10^{10}\ g$$

Ответ: а) $F_x = -2F_0\quad F_y = 0\quad a_{max} = 2{,}3\cdot10^{10}\ g$

б) $F_x = \pm F_0\quad F_y = 0$

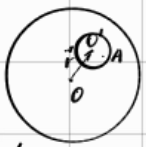
в) $F_x = 0\quad F_y = \pm F_0\quad F_0 \simeq 7\cdot10^{-15}\ Н$

1.22

Дано:
ρ ; r
Найти: E

Решение:

Представим шарик электрических противоположных зарядов с плотн. $+\rho$ и $-\rho$
Поле внутри – сумма полей двух заряженных шаров. Возьмем точку A внутри малого шара:



$$\vec{E} = \vec{E}_{O'} + \vec{E}_{O} = \frac{4\pi}{3}\left(\rho\,\overrightarrow{OA} + (-\rho)\,\overrightarrow{O'A}\right) = \frac{4\pi}{3}\rho\left(\overrightarrow{OA} - \overrightarrow{O'A}\right) = \frac{4\pi}{3}\rho\vec{r}$$

Ответ: $\frac{4}{3}\pi\rho\vec{r}$

1.23

Дано: $R; E_0$
Найти: $\sigma(\theta); E$

Решение

Сделаем сферу вдоль вект. оси. $\exists\, OO = \vec{r}$. В области шарика сферу заряда компенсируются. Толщина "сдвиг." слоя: $r = r\cos\theta$. Тогда $\sigma(\theta) = \rho r\cos\theta$

$$E = \frac{4}{3}\pi\rho r \Rightarrow \rho = \frac{3E_0}{4\pi r} \qquad \sigma(\theta) = \frac{3E_0}{4\pi}\cos\theta$$

Д-ть дип. момент $\vec{p} = q\vec{r} = \rho\frac{4\pi}{3}R^3\vec{r} = \frac{3\vec{E}_0}{4\pi}\frac{4\pi}{3}R^3 = R^3\vec{E}_0$

Ответ: $\sigma(\theta) = \frac{3E_0}{4\pi}\cos\theta$
$\vec{p} = R^3\vec{E}_0$

1.17

Дано:
$R = 10^{-8}$ см
$e = 1{,}6\cdot10^{-12}$ эрг
Найти: $\rho(r)$; ν

Решение

$m\ddot{r} = -kr \Rightarrow \omega = \sqrt{\frac{k}{m}}$

$\omega = \text{const} \Rightarrow k = \text{const} \qquad F = kr = eE \Rightarrow E = \frac{kr}{e}$

$\operatorname{div}\vec{E} = 4\pi\rho(r)$
$\operatorname{div}\vec{E} = \frac{1}{r^2}\frac{d(r^2E)}{dr} = \frac{2E}{r} + \frac{k}{e} \Rightarrow 4\pi\rho(r) = \frac{2E}{r} + \frac{k}{e} = \frac{2k}{e} + \frac{k}{e} = \frac{3k}{e} \Rightarrow \rho = \frac{3k}{4\pi e}$

$$e = \int_V \rho\,dV = \int_0^R 4\pi r^2\rho\,dr = \frac{4}{3}\pi R^3\frac{3k}{4\pi e} = \frac{R^3k}{e} \Rightarrow k = \frac{e^2}{R^3}$$

$$\rho = \frac{3k}{4\pi e} = \frac{3e}{4\pi R^3} =$$

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{e^2}{mR^3}} \qquad \nu = \frac{1}{2\pi}\sqrt{\frac{e^2}{mR^3}} = 2{,}5\cdot10^{15}\ \text{Гц}$$

1.16

Дано:
d ; $\rho = \rho_0\frac{x}{d}$; m; $\vec{P}$
Найти: T

Решение

$\operatorname{div}\vec{E} = 4\pi\rho = 4\pi\rho_0\frac{x}{d} \Rightarrow \frac{\partial E}{\partial x} = 4\pi\rho_0\frac{x}{d} \Rightarrow E(x) = 2\pi\rho_0\frac{x^2}{d} + C$
$E(0) = 0 \Rightarrow C = -2\pi\rho_0\frac{d^2}{d}$

$\vec{P} = q\vec{l}$

$$F = qE = q\,2\pi\frac{\rho_0}{d}\left(\left(\left(\frac{l}{2} - x\right)^2 - d^2\right) - \left(\left(\frac{l}{2} + x\right)^2 - d^2\right)\right) = -4\pi q\frac{\rho_0}{d}lx = -4\pi\frac{\rho_0}{d}px$$

$m\ddot{x} = F \qquad m\ddot{x} + 4\pi\frac{\rho_0}{d}px = 0 \qquad \ddot{x} + \frac{4\pi\rho_0 P}{md}x = 0 \qquad T = \frac{2\pi}{\omega} = \pi\sqrt{\frac{md}{\pi\rho_0 P}} = \sqrt{\frac{\pi md}{\rho_0 P}}$

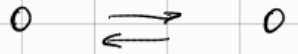
Т 1.2

Дано:
l; q; P

Найти: F

Решение:

а)

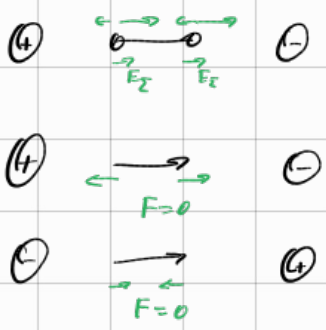


1. одноим. зар.

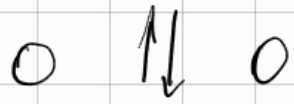


$$|F| = \left|P\frac{\partial E}{\partial x}\right| = \left|P\frac{\partial}{\partial x}\left(\frac{3P}{(l/2)^3} - \frac{P}{(l/4)^3}\right)\right| = \frac{48P^2}{l^4}$$

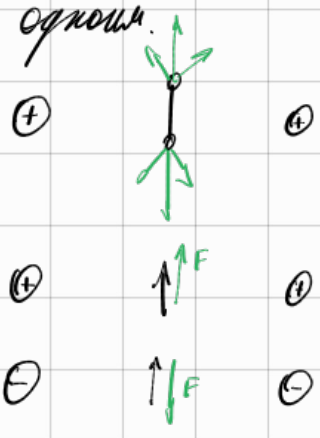
2. разноим.



б)

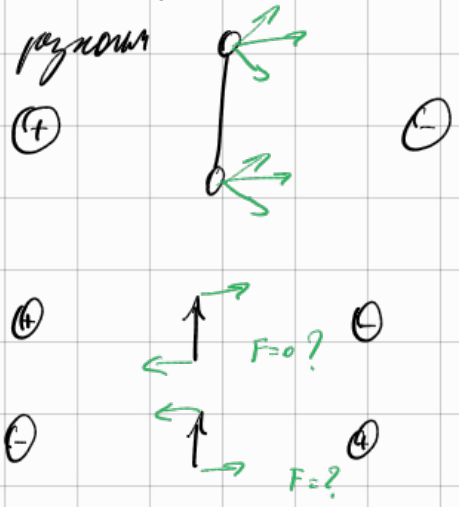


1. одноим.



$$|F| = \left|P\frac{\partial E}{\partial x}\right| = \left|P\frac{\partial}{\partial x}\left(-\frac{P}{(l/2)^3}\right)\right| = \frac{24P^2}{l^4}$$

2. разноим.



1.21

Дано:

E_0

Найти: $\rho(r)$

Решение

1) $\operatorname{div} \vec{E} = 4\pi\rho$

$\operatorname{div} \vec{E} = \frac{1}{r^2} \frac{\partial(r^2 E)}{\partial r} = \frac{E_0}{r^2} 2r = \frac{2E_0}{r}$

$\rho = \frac{E_0}{2\pi r}$